Temperature probe

DETECT | PICO | DS18B20 | DIGITAL

What does it do?

A **DS18B20** measures temperature and sends a digital reading in degrees Celsius.

This probe has a sealed metal tip on a cable with three wires. Check the supplier's pinout; wire colours can vary.

Only the sealed probe goes in water. Keep the board and all wire connections dry.

Uses



Aquarium temperature alarm

DETECT Measure tank temperature → **DECIDE** Is it too cold? → **DO** Show a warning



Penguin bath alarm

DETECT Measure bath temperature → **DECIDE** Is it too warm? → **DO** Light a warning LED



VDD	DQ	GND	4.7 kΩ
Power from 3V3	Digital data	Ground	Resistor from DQ to 3V3

Raspberry Pi Foundation

Code

Raspberry Pi Pico

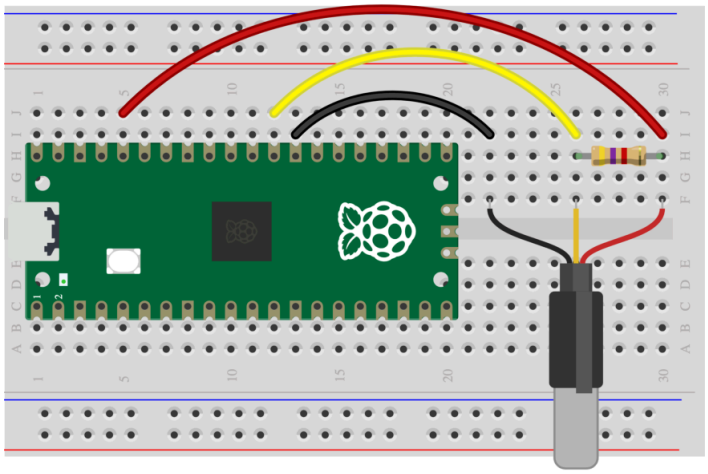
```
from machine import Pin
import onewire, ds18x20
from time import sleep_ms

bus = onewire.OneWire(Pin(22))
sensor = ds18x20.DS18X20(bus)
probes = sensor.scan()
assert probes, "Check probe wiring"
sensor.convert_temp()
sleep_ms(750)
temperature = sensor.read_temp(probes[0])
print(temperature)
```

Repeat the conversion and 750 ms wait for each new reading.

Wiring

PICO



VDD → 3V3 OUT (pin 36); **DQ** → GP22 (pin 29); **GND** → GND (pin 28). **4.7 kΩ** between DQ and 3V3.